SMART INDIA HACKATHON 2025

AUTOMATIC FABRIC INSPECTION SYSTEM FOR TEXTILE INDUSTRY

- **Problem Statement ID –** SIH25118
- **Problem Statement Title –** Student Innovation Automatic Fabric Inspection System for Textile Industry
- **Theme –** Smart Automation
- **PS Category –** Hardware
- **Team ID - 75056**
- **Team Name -** FabSpector

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AUTOMATIC FABRIC INSPECTION SYSTEM FOR TEXTILE INDUSTRY

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📝 Proposed Solution

Trained deep learning model identify defects captured fabric images.

System sends defected data to a **Cloud** for monitoring, logging, and quality analysis.

System trigger **motor control unit** to stop the machine, marking the defective section for corrective action.

Provides **real-time alerts, defect images, and production statistics** for supervisors

📌 Problem Statement

Nowadays Industries are facing many challenges during manual fabric inspection system. Manual errors will occur during inspecting defects like **uneven dying, oil stain, yarn defect, large and tiny holes** etc.

MANUAL INSPECTION METHODS IN TEXTILE INDUSTRIES

HOLES

OIL STAINS

✅ How it Addresses the Problem

- Spot hidden defects reducing manual errors.
- Works in real time without delays in production.
- Every defect is saved with images and details, helping industries analyze patterns and improve quality over time.
- Prevents **defective fabric** from reaching customers, reducing losses, complaints, and rework costs.
- Can be added to many production lines without hiring extra manpower.
- Allows workers to focus on skilled tasks like **quality control, maintenance, and process improvement.**

💡 Innovation and Uniqueness

- Combines real-time edge detection with powerful cloud-based analysis, balancing speed and intelligence and **directly control fabric rollers** to stop at the defective portion.
- Generates **defect reports and analysis** and the system can adapt to different defect categories by retraining the **AI model**.
- **AI** does the repetitive inspection validate critical defects, ensuring balance between automation and expertise.
- Integrates the system with existing available machines and Reducing **fabric waste and rework**

TECHNICAL APPROACH

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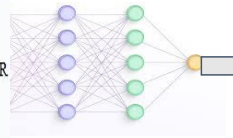
IEEE YESIST12 2025 WINNER UNDER FET100 TRACK
"AUTOMATIC FABRIC INSPECTION SYSTEM FOR TEXTILE INDUSTRY"

WORKFLOW OF PROPOSED PROTOTYPE

Dataset collection & Model Training

Model deployment to CAM module for real time interface

ric feeding & Motion control



Defect Detection using CNN

Real time implementation for image capturing

Motion control using motor drive for controlled algorithm

Storage of detected image in cloud using IOT integration

Prototype survey, analysis, working and explanation video:
https://drive.google.com/drive/folders/1uBCRgxcFI5DfS6Bi2pdIJtUXnFhkYPX-?usp=sharing

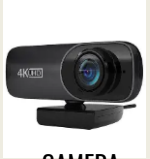
Multi-model concept for fabric inspection on both edge and cloud:

- ⚡ **Real-time defect detection on edge → MobileNetV3** Small light, fast model that quickly flag fabric defects
- 🎯 **High-accuracy validation on cloud → EfficientNetB0** heavier model runs in the cloud/PC
- 🔄 **Fallback for reliability → ShuffleNetV2** ultra-light model works smoothly when edge devices have limited power or resources

FIELD VISIT IN SHAKTHI KNITTING COMPANY

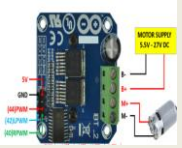
TECHNOLOGY STACK

HARDWARE



CAMERA

RASPBERRY PI



bts7960 motor driver and 12v dc motor

SOFTWARE

PYTHON
SEABORN
TENSORFLOW
KEARAS
MATPLOTLIB
NUMPY
SCIKIT LEARN
OPEN CV
PILLOW

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FEASIBILITY AND VIABILITY

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FEASIBILITY

Technical: Highly feasible using affordable, off-the-shelf hardware (Raspberry Pi, cameras) and mature, open-source software (OpenCV, TensorFlow).

Operational: Easily integrates into existing factory workflows with a simple user dashboard, requiring minimal staff training and maintenance.

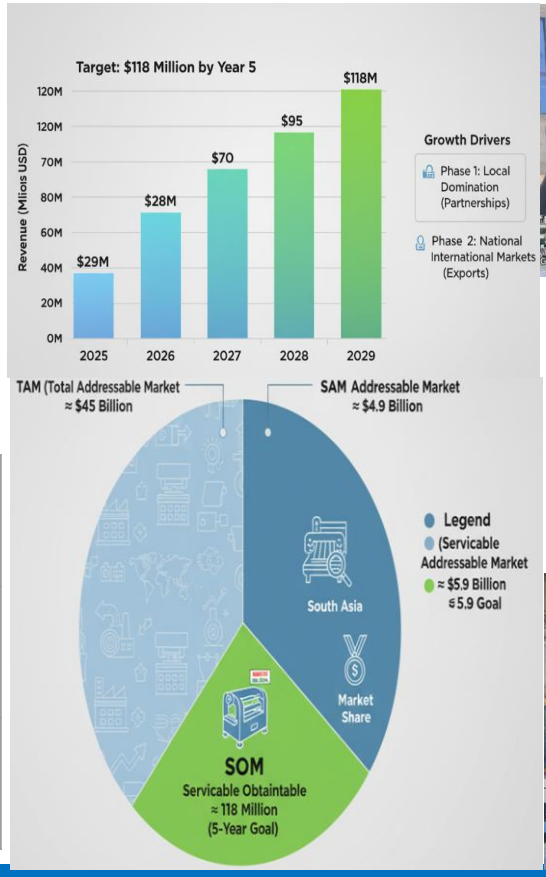
Economic: Offers a strong and rapid Return on Investment (ROI) due to low initial costs and significant savings from reduced labor, increased speed, and less material waste.

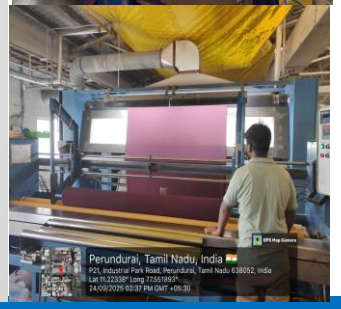
POTENTIAL CHALLENGES

- **Lack of a Diverse, Labeled Dataset:** making robust AI model training extremely difficult.
- **Environmental Interference & Variability:** Fluctuating ambient light, dust, and high-speed fabric vibrations can severely degrade camera image quality, leading to inaccurate defect detection.
- **Complexity of Fabrics and Defects:** Model may fail to identify subtle defects on complex patterns, dark fabrics, or textured surfaces.

STRATEGIES

- **Create a Proprietary Dataset:** data-sharing partnerships with local mills creating an unmatched, proprietary training asset.
- **Engineer a Controlled Inspection Module:** physical enclosure for the camera with integrated lighting eliminates external light
- **Hybrid AI Approach:** deep-learning model to classify known defects, while a secondary anomaly detector (trained on flawless fabric) flags new or unseen imperfections, even in complex patterns.





MARKETING STRATEGIES

Total Addressable Market (TAM)	Global fabric production, labor costs, cost of quality failures.	$45 - $50 Billion
Serviceable Addressable Market (SAM)	Go-to-market strategy, product specifications, target customer.	~$5.88 Billion
Serviceable Obtainable Market (SOM)	Competition, sales execution, product USP, pricing.	~$117.6 Million (5-Year Goal)

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IMPACT AND BENEFITS

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BENEFITS:

- Spots even the **smallest fabric flaws**—defects that might easily be missed by the human eye.

- **Reliable Consistency** doesn't tire or lose focus, ensuring each inspection remains just as thorough as the one before.

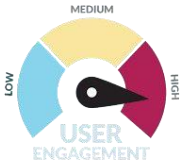
- **Saves Time** which take hours to do manually now gets done in minutes, keeping things running smoothly.

- **Boosts Productivity** by more efficient inspections, your production line stays on track.
- **Cuts Costs** by reduced waste, fewer mistakes, and lower labor requirements lead to significant savings.

- **Better Quality Control** providing more defect-free fabric reaches customers, enhancing the overall product quality.
- **Competitive Edge** by adhering to global standards and consistently delivering high quality, textile companies can build trust and stay ahead in the market.

IMPACTS:

- **Real-Time Quality Checks** Problems are addressed before they worsen because flaws are identified immediately during production.



- **Efficient Operations** provides entire process is more cost-effective when expenses are reduced and resources are used more wisely.

- **Faster Output Inspection** increases production rather than slows it down.

- **Increased Customer Confidence** by reputation and loyalty are fostered by consistent quality.

- **Encourages Smart Manufacturing** by assists sectors in moving toward automation and workflows that are prepared for the future.

- **Decreased Human Errors** lessens the possibility of errors and weariness that accompany manual inspection.

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RESEARCH AND REFERENCES

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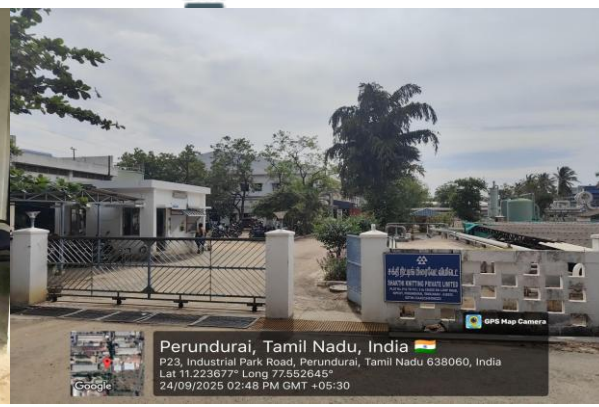

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prototype survey, analysis, working and explanation video:
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